

June 19, 2026

Checks-in, and Balances: a Matter of Time

Definitions

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Contents

I.	Schedule, seats & passengers	1
II.	Load, existence & entry	2
III.	Ground handling	2
IV.	Check-in desks, service & queueing	2

Living registry per Ground Rule 2. Updated as relevant as the model is built.

I Schedule, seats & passengers

Term	Definition
Foundation schedule	The canonical one-row-per-flight departure schedule for the operating day, held in <code>asimplified.csv</code> (164 flights after the B4 443 omission). The input to Step 1.
Corpus	The cleaned schedule table of the six author-defined variables (<code>flight</code> , <code>n</code> , <code>route</code> , <code>dest</code> , <code>cap</code> , <code>dep</code>), one row per flight ordered by <code>n</code> . Held as the <code>corpus</code> tab of <code>asimplified.xlsx</code> .
Seat	The seat serial (variable <code>seat</code>): one per occupiable seat, continuous over all flights in departure order. The seat-level master is <code>seats.csv</code> .
Seat universe	The maximal set of occupiable seats, one per seat, of size $N = \text{sum cap}$. A passenger is an occupied seat; the realised count per flight is $\text{round}(\text{LF} \times \text{cap})$.
Passenger (pax)	A realised passenger: an occupied seat (<code>exists = 1</code>). Per-flight count <code>pax</code> = $\text{round}(\text{LF} \times \text{cap})$.
Operating day (“Day 7”)	The single schedule day modelled: 21-06-2026. All <code>dep</code> times fall within this day.
SOBT	Scheduled Off-Block Time - the scheduled time a flight pushes back from its stand. Used as the variable <code>dep</code> .
Flight designator	Carrier code plus flight number identifying a scheduled service (variable <code>flight</code>).
Route / Routing	The ordered sequence of airports for a service, origin DEL (variable <code>route</code>).
Destination	The final airport of the route (variable <code>dest</code> , 3-letter IATA).
IATA code	Three-letter airport identifier (e.g. DEL, SFO).
Capacity	Number of seats offered on a flight (variable <code>cap</code>).
Many-to-one assignment	A mapping in which each seat is assigned to exactly one flight while a flight carries many seats.

Term	Definition
B4 443 omission	Flight B4 443 (Beond, DEL-MLE) is excluded from the analysis: it is not a regularly scheduled service and the source lists neither a Ground Handler nor a Capacity for it. See Methodology, Data Note.

II Load, existence & entry

Term	Definition
Load factor (LF)	Assumed fraction of seats occupied; realised passengers per flight = $\text{round}(\text{LF} \times \text{cap})$. Varied across scenarios (e.g. 0.90).
Existence (exists)	$\text{exists}(\text{seat})$ in $\{0, 1\}$: 1 if a seat is occupied (a pax), else 0. Unoccupied seats have no entry time.
Check-in window	Opens at dep - 0400 (four hours before departure) and closes at dep - 0100 (one hour before). Equals $[\text{checkinopen}, \text{checkinclose}]$ in the data.
Entry time	$\text{entry}(\text{pax})$: the time at which a passenger (pax) enters the check-in area; defined for each pax .
Entry-time model (m)	A distribution function over entry times. Different models distribute $\text{entry}(\text{pax})$ in different ways; passenger counts vary with LF.

III Ground handling

Term	Definition
GHA (Ground Handling Agent)	The agent providing ground/check-in handling for a flight. Values in the schedule: AISATS, BFS, Ramp 360, AIESL.
AISATS	AI-SATS, a ground-handling joint venture. Treated as a distinct GHA.
AIESL	Air India Engineering Services Ltd - treated as a Ground Handler independent of and not merged with AISATS .

IV Check-in desks, service & queueing

Term	Definition
Check-in desk (“point”)	A single check-in position (variable k). An airline has $D(a, t)$ desks open at time t ; this can change over time.
Desk operating window	Constraint that desks for a flight may be open only within $[dep - 0400, dep - 0100]$. In the data this equals $[checkinopen, checkinclose]$ for every flight.
Processing speed	Service time per passenger, τ (seconds per pax), or equivalently a service rate $\mu = 1/\tau$ (pax per unit time).
Eligibility ($E(p)$)	The set of desks a passenger may use, determined by <code>class</code> , <code>loyaltystatus</code> , and <code>fastforward</code> . (Mapping to be formalised.)
<code>loyaltystatus</code>	Loyalty tier 0-5: none, silver, gold, platinum, 1K-equivalent, GS-equivalent.
<code>class</code>	Cabin class 1-4: economy, premium economy, business, first.
<code>fastforward</code>	Binary $\{0,1\}$: whether a passenger has purchased queue priority (e.g. IndiGo).
Wait time (w)	Time a passenger waits before service begins. For a single eligible desk: (passengers ahead) $\times \tau$ minus time already elapsed on the passenger currently in service. With several eligible desks, the passenger is assumed to choose the desk with least wait.
Rational counter choice	Assumption that a passenger joins the eligible open desk offering the least wait time.
OMDA Target	Benchmark service level the model should ideally meet: ≤ 5 min wait for any business-class passenger and ≤ 20 min for any economy passenger. (Sidenote / benchmark, not a model input.)